

## B: Be Strong

Source file name: `be.c`, `be.cpp`, `be.java`, or `be.py`

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A *prefix* of a string is a substring that occurs at the beginning. If a string has size  $N \geq 0$ , then it has exactly  $N + 1$  prefixes (the empty string is a prefix of any string). Given a collection of strings, its *strong prefix* is the longest prefix common to all strings in the collection. Any difference between lowercase and uppercase characters is considered immaterial.

For example, `te` is the strong prefix for the collection

`tequila`      `tEnnessee`      `Telephone`      `tetris`

The empty string is the strong prefix for the collection

`hello`      `world`

In this problem, you are asked to compute the strong prefix of a collection of words.

### Input

The input consists of several test cases. Each test case begins with a line containing a number  $M$  ( $0 \leq M \leq 5\,000$ ) denoting the number for words in the collection. Then,  $M$  lines follow, each containing a string  $W$  ( $1 \leq |W| \leq 200$ ) made from English lowercase and uppercase characters. The end of the input is given with  $M = 0$ , which should not be processed.

*The input must be read from standard input.*

### Output

For each test case, output a single line with the strong prefix of the collection of  $M$  words where the case is immaterial. The output can only contain English lowercase characters. If the strong prefix is empty, output a star `*`.

*The output must be written to standard output.*

| Sample Input                                                                                  | Sample Output |
|-----------------------------------------------------------------------------------------------|---------------|
| 4<br>tequila<br>tEnnessee<br>Telephone<br>tetris<br>2<br>hello<br>world<br>2<br>De<br>dE<br>0 | te<br>*<br>de |